Uniform pool training · transfer gap across all checkpoints

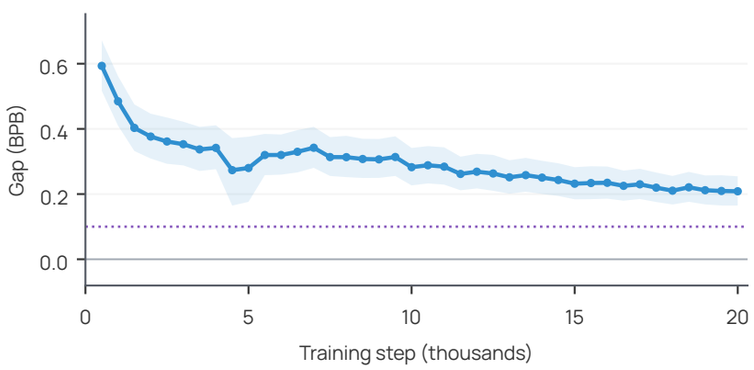
Transferred BPB − trained canonical BPB at the same checkpoint · lower is better

Transfer gap · +0.10 BPB weakness threshold · 95% source-document cluster interval

Chinese35 · BPE 48k / Chinese

96 passages / 96 docs · 63% fewer tokens · original panel

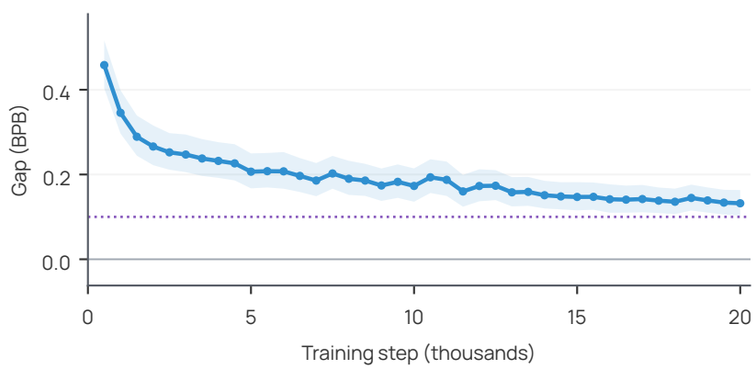
Tokenizer absent from training pool



Japanese35 · BPE 48k / Japanese

96 passages / 96 docs · 72% fewer tokens · original panel

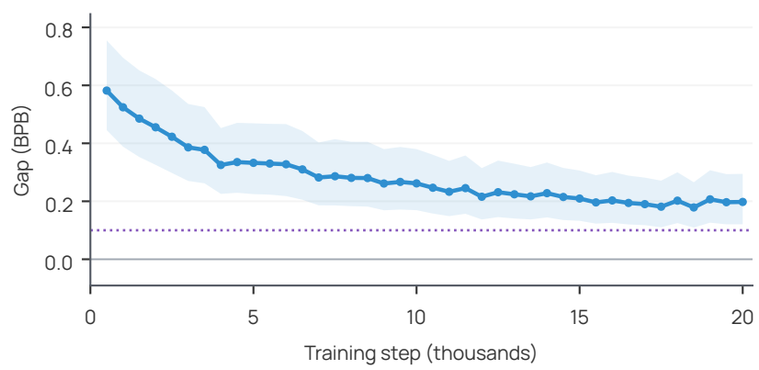
Tokenizer absent from training pool



Base BPE 64k / medical English

96 passages / 96 docs · 23% fewer tokens · original panel

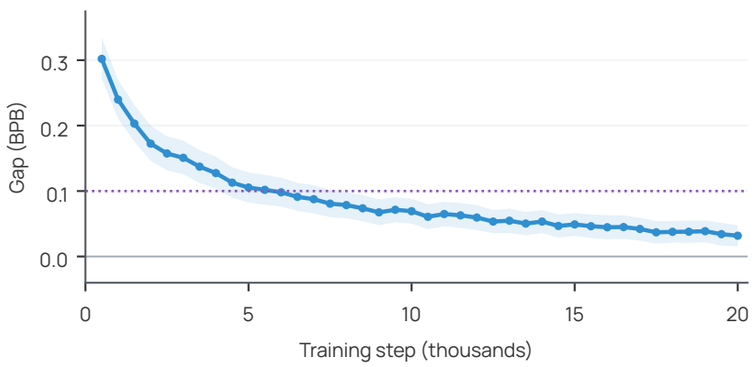
Tokenizer absent from training pool



Pool 416 · WordPiece 102k / medical

256 passages / 102 docs · 42% fewer tokens · later passages

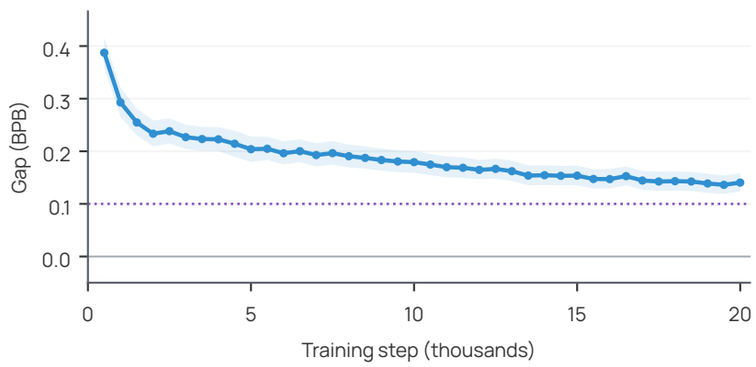
In pool: sampled 23 times by 20k



Pool 822 · Unigram 123k / Japanese

256 passages / 256 docs · 71% fewer tokens

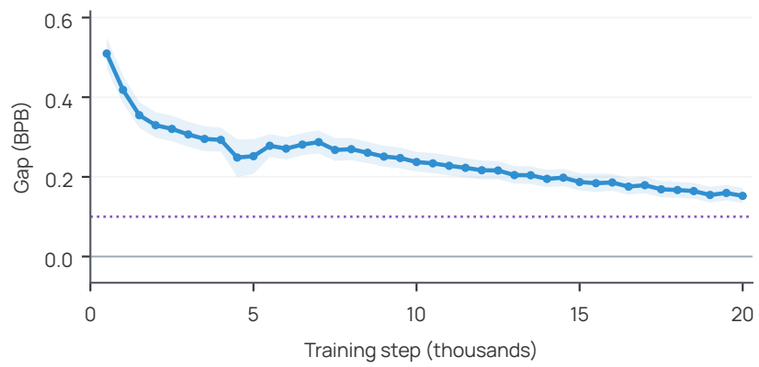
In pool: sampled 16 times by 20k



Pool 616 · BPE 115k / Chinese

256 passages / 256 docs · 57% fewer tokens

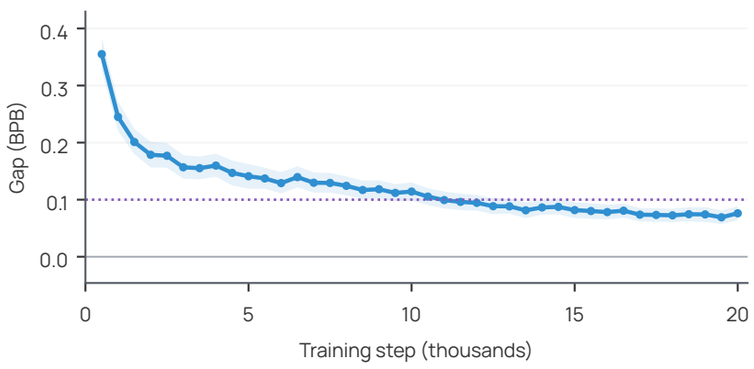
In pool: sampled 26 times by 20k



StarCoder2 · 49k / Japanese

256 passages / 256 docs · 45% fewer tokens

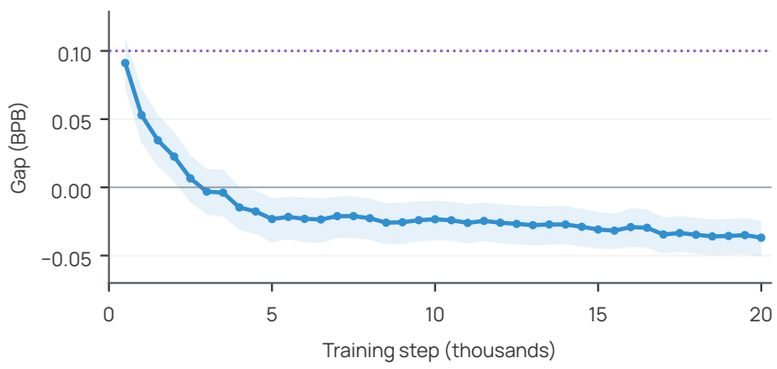
Tokenizer absent from training pool



WordPiece 150k† / medical English

256 passages / 102 docs · 16% fewer tokens · later passages

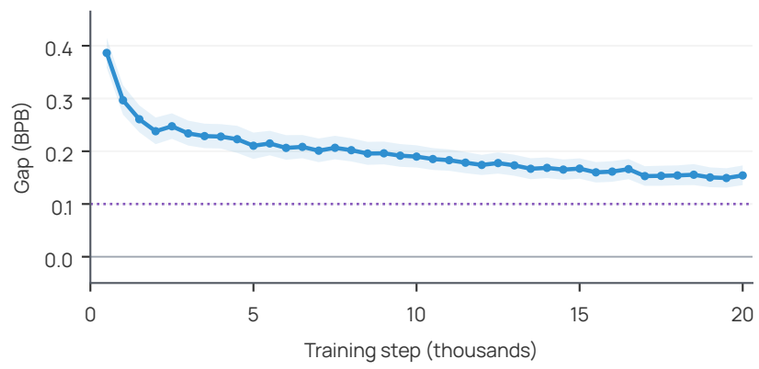
† Vocabulary outside training range



Unigram 150k† / Japanese

256 passages / 256 docs · 73% fewer tokens

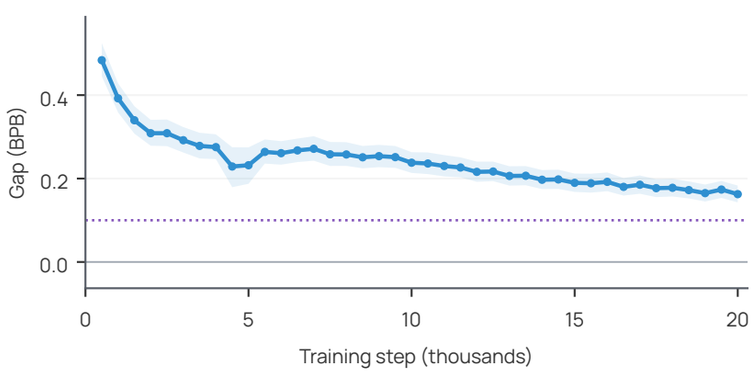
† Vocabulary outside training range



Qwen2.5-Coder · 152k† / Chinese

256 passages / 256 docs · 61% fewer tokens

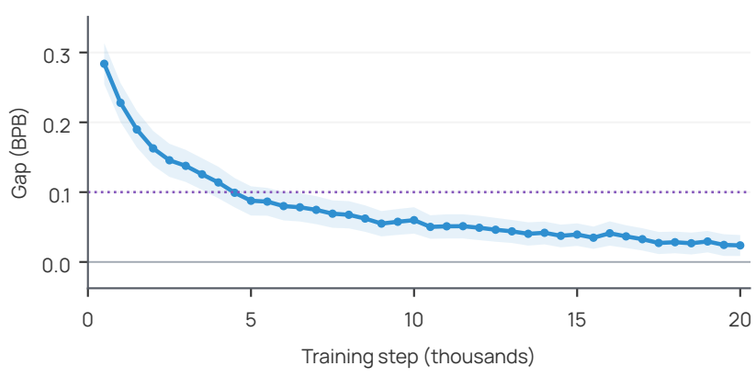
† Vocabulary outside training range



Pool 635 · BPE 120k / medical

256 passages / 102 docs · 42% fewer tokens · later passages

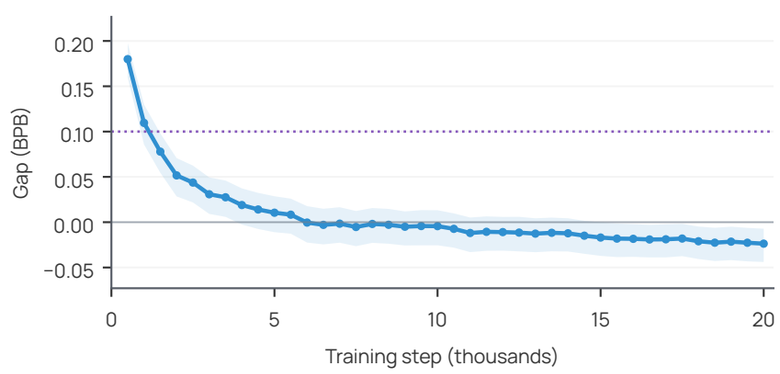
In pool: sampled 20 times by 20k



Base BPE 48k / English control

256 passages / 256 docs · 12% fewer tokens

Tokenizer absent from training pool



32 canonical continuation tokens · BOS + prefix + continuation ≤128 positions · canonical output head fixed

Original panels: canonical prefix ≤95 tokens. New panels: ≤64. Medical later passages cluster by source article.

† 150k/152k vocabularies exceed the training pool's 32k–128k range. Curves describe one training run.